

Alan Anto

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EDUCATION

- Brandenburg University of Technology** Cottbus, Germany
MSc.in Artificial Intelligence, Grade: 2.2/5
Courses: Cognitive Systems, Image processing, Neural Networks, Explainable Machine Learning
Apr 2025 - Apr 2027
- APJ Abdul Kalam Technological University** Kerala, India
B.Tech(Hons.)in Artificial Intelligence and Data Science, Grade: 9.1/10
Courses: Data Structures, Data Science, Artificial Intelligence, Machine Learning, NLP, Database systems
Dec 2020 - Jun 2024

SKILLS SUMMARY

- Languages:** Python, SQL, C++
- GenAI Framework:** LLM, Transformers, RAG, Prompt Engineering, Qlora, Hugging Face
- Agentic AI Frameworks:** LangChain, LangGraph, BeeAI, crewAI, Google ADK
- ML & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, Numpy
- Databases:** MySQL, VectorDB, GraphDB
- Cloud & DevOps:** AWS, Azure, Docker, Git, Gradio, REST API, IBM Watsonx
- Soft Skills:** Leadership, Event Management, Documentation, Time Management, MS office

EXPERIENCE

- Indian Institute of Information Technology** Kerala, India
AI Engineering Intern Jun 2023 - Aug 2023
 - Achieved real-time inference on edge devices with speeds of **30 FPS** by optimizing **YOLOv8** object detection models for limited hardware.
 - Boosted model robustness by **15%** against environmental variations using an automated **OpenCV pipeline** to augment over **5,000** training images with noise and rotation.
 - Benchmarked multiple **computer vision** algorithms, as measured by precision-recall under edge constraints, by designing a systematic evaluation framework optimized for deployment reliability.

PROJECTS

- AutoRAG Intelligence** | Neo4j, LangGraph, ChromaDB, FastAPI, Gradio, Ollama, Docker LINK
 - Engineered a production-grade **hybrid RAG** system for automotive intelligence that reduced hallucinations by **40%** and optimized **routing efficiency** through a 10-node **LangGraph agent pipeline** using **Neo4j** and **ChromaDB**.
 - Built a secure, full-stack Python application achieving **100%** data privacy and **40%** faster response delivery by orchestrating a **Asnyc FastAPI** backend and **Gradio** frontend with local LLM inference.
 - Developed an end-to-end **data ingestion and indexing pipeline**, achieving **faster retrieval** and a **25%** increase in accuracy validated via **RAGAS**, by implementing custom automotive **entity extraction** and a **hybrid indexing** schema in Neo4j.
- Agentic Code-Mate** | Google ADK, Groq, Streamlit, pytest, llama LINK
 - Engineered an autonomous, **multi-agent AI coding assistant** that accelerated developer workflows by **35%** and cut cloud infrastructure expenses by utilizing the **Google ADK** framework and the **Groq API**.
 - Developed an **intelligent orchestration** system that achieved a **95%** intent-routing accuracy and **real-time streaming** capability by designing an **Orchestrator agent** to dynamically manage high-capacity LLMs.
 - Built a **multi-file ingestion** and automated code-refactoring engine that expanded unit test coverage by **40%** and removed critical security vulnerabilities through **specialized sub-agents**.
- Autonomous Multi-Agent Travel System** | BeeAI, Custom Tools, Fast API, React, Async LINK
 - Engineered a **multi-agent travel planner**, achieving **100%** automated data synthesis by architecting a hierarchical **BeeAI framework** with specialized expert agents and **handoff protocols**.
 - Reduced user-perceived latency by **65%** for reasoning tasks by developing an **asynchronous FastAPI** backend that streams real-time agent trajectories and tool events via SSE to a **React UI**.
 - Enhanced LLM reliability by implementing strict execution **guardrails** using **BeeAI RequirementAgents** to enforce conditional tool logic and maintain state during complex agent **handoffs**.

PUBLICATIONS

- Autonomous Wheeled Robot for Intelligent Supermarket Restocking** | LINK Jun 2025
IEEE 2nd International Conference on Trends in Engineering Systems and Technologies (ICTEST)

CERTIFICATIONS

- IBM RAG and Agentic AI Professional Certificate** | CERTIFICATE Dec 2025
- AWS Generative AI for Developers Professional Certificate** | CERTIFICATE Oct 2025
- Microsoft Azure AI Essentials Professional Certificate** | CERTIFICATE Aug 2025
- IBM AI Engineering Professional Certificate** | CERTIFICATE Jul 2025
- Docker Foundations Professional Certificate** | CERTIFICATE Nov 2025

LANGUAGES

- German:** B1 (Currently pursuing B2)
- English:** C1 (Professional Working Proficiency)